Chapter 2 — Softmax and temperature

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Type: theory + notebook
Ordering: theory-first
Depends on: 1 (scaled dot-product attention)
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2.1 Motivation

Attention scores s_{ij} are arbitrary reals. To turn them into usable attention weights we need a map $\mathbb{R}^n \to \Delta^{n-1}$ (the probability simplex) that is non-negative, sum-to-one, differentiable, monotone in each coordinate, and — ideally — the exponential-family canonical link. Softmax satisfies all of these and is the unique such map up to scaling. Its shape parameter, temperature, controls how selectively attention concentrates. Understanding softmax deeply is understanding attention's information-retrieval dynamics.

2.2 Storyline

Softmax does one job with four useful consequences.

The job. Given logits $s \in \mathbb{R}^n$, softmax emits the Gibbs distribution $p_i = e^{s_i/T}/Z$ on $\{1, \dots, n\}$, with temperature $T > 0$ controlling concentration.

Useful consequence 1: shift invariance. Adding a constant to every logit leaves the distribution unchanged. This is why the standard numerical implementation subtracts $\max_i s_i$ before exponentiating — it prevents overflow without affecting the answer.

Useful consequence 2: smoothness. Softmax is C^∞, with a clean Jacobian $\partial p_i/\partial s_j = p_i(\delta_{ij} - p_j)/T$. This enables gradient flow to the upstream Q, K projections. Its failure at saturation (Lemma 3.3 preview) is the central reason scaling matters.

Useful consequence 3: temperature controls entropy. As $T \to 0^+$, softmax approaches a one-hot at $\arg\max s$; as $T \to \infty$, it approaches uniform. The entropy of p is a smooth monotone function of T for fixed s. This is the mechanism's information-control knob.

Useful consequence 4: log-sum-exp as a soft max. $\mathrm{LSE}(s) = \log \sum_i e^{s_i}$ is a smooth upper bound on $\max_i s_i$, and $\nabla \mathrm{LSE} = \mathrm{softmax}$. This is why "softmax" is called that — it is the gradient of a smooth-max function.

Temperature in attention. In scaled dot-product attention the $1/\sqrt{d_k}$ factor is exactly an effective temperature: $\mathrm{softmax}(QK^\top/\sqrt{d_k}) = \mathrm{softmax}_{\sqrt{d_k}}(QK^\top)$. Chapter 3 derives why this particular temperature is the right one; Chapter 2 is the tool-building that makes Chapter 3 possible.

Regime notes. Softmax itself is stateless — no train / val / inference asymmetry in the operator. Temperature, however, can differ across regimes: - Training uses whatever temperature is built into the architecture (typically $\sqrt{d_k}$ for attention, 1 for the output layer). - **Inference-time sampling** from a language model often uses a different temperature (sampling temperature) on the output softmax — concentrating or broadening the next-token distribution independently of how the model was trained. This is a post-training knob. - Masking (causal or attention masks) is applied additively to the pre-softmax scores, usually by adding $-\infty$ to disallowed positions, producing zero weight there after softmax.

2.3 Rigorous core

Definition 2.1 (Softmax with temperature). For $s \in \mathbb{R}^n$ and $T > 0$,

$$\text{softmax}_T(s)_i = \frac{e^{s_i/T}}{\sum_{j=1}^n e^{s_j/T}}.$$

We write $\text{softmax}(s) := \text{softmax}_1(s)$ for the unit-temperature default.

Proposition 2.2 (Shift invariance). For any $c \in \mathbb{R}$, $\text{softmax}_T(s + c\mathbf{1}) = \text{softmax}_T(s)$. (The constant $e^{c/T}$ factors out of numerator and denominator.)

Proposition 2.3 (Jacobian). Let $p = \text{softmax}(s)$. Then

$$\frac{\partial p_i}{\partial s_j} = p_i(\delta_{ij} - p_j).$$

For softmax_T, the Jacobian is $\frac{1}{T} p_i(\delta_{ij} - p_j)$. Quotient rule, case-split on $i = j$.

Corollary 2.3.1 (Mass conservation). The rows and columns of the Jacobian each sum to zero: $\sum_i \partial p_i / \partial s_j = 0$. Why: softmax maps to the simplex, so $\sum_i p_i = 1$ is constant; any perturbation in s_j preserves this.

Proposition 2.4 (Entropy bounds). For $p = \text{softmax}(s)$, $H(p) = -\sum_i p_i \log p_i \in [0, \log n]$. The upper bound is attained iff all s_i are equal; the lower bound is approached as the pre-softmax margin grows without bound. Upper: Jensen / Lagrange. Lower: entropy non-negative on any distribution.

Proposition 2.5 (Saturation margin). For $s = (0, \Delta, 0, \dots, 0) \in \mathbb{R}^n$ (winner at index 2 with margin Δ), the winning probability is

$$p_2 = \frac{e^\Delta}{e^\Delta + (n-1)} = \frac{1}{1 + (n-1)e^{-\Delta}}.$$

To achieve $p_2 > 1 - \epsilon$ requires $\Delta > \log((n-1)/\epsilon)$ to leading order.

Corollary 2.5.1. Margin requirements are logarithmic in n — doubling the number of keys costs one additional nat of margin. This is why attention over long contexts is not hopeless despite the n-way competition; the cost to distinguish a clear winner from n distractors grows only as $\log n$.

Proposition 2.6 (Log-sum-exp). Let $\text{LSE}(s) = \log \sum_i e^{s_i}$. Then:

(a) $\nabla \text{LSE}(s) = \text{softmax}(s)$;
(b) $\max_i s_i \le \text{LSE}(s) \le \max_i s_i + \log n$;
(c) LSE is convex.

Sketches: (a) direct $\partial_{s_i} \text{LSE} = p_i$. (b) sandwich $e^m \le \sum_i e^{s_i} \le ne^m$. (c) Hessian is the softmax Jacobian, which is PSD as the covariance of the categorical p — Exercise 2.1.

Proposition 2.7 (Temperature as inverse logit scaling). $\text{softmax}_T(s) = \text{softmax}(s/T)$. Therefore the $1/\sqrt{d_k}$ factor in attention corresponds to effective temperature $T_{\text{eff}} = \sqrt{d_k}$.

2.4 Numerical example — shift invariance and saturation by hand

Shift invariance. $\text{softmax}((1, 2, 3))$ vs. $\text{softmax}((101, 102, 103))$: identical. The second form overflows a naive implementation; subtracting 103 first recovers the first form.

Hand-computed:

$$\text{softmax}((1,2,3)) = \frac{1}{e+e^2+e^3}(e, e^2, e^3) \approx (0.090, 0.245, 0.665).$$

Saturation by margin. For $n = 10$ and $\Delta \in \{1, 3, 5, 10\}$ the winning probability $p_{\text{win}} = e^{\Delta}/(e^{\Delta}+9)$ is:

Δ	p_{win}
1	≈ 0.252
3	≈ 0.691
5	≈ 0.943
10	≈ 0.9996

Margin $\Delta \approx \log(n/\epsilon)$ is where p_{win} crosses $1-\epsilon$: for $n = 10$, $\epsilon = 0.01$, that's $\Delta \approx 6.9$ — the table above brackets it.

LSE vs. max. For the same vector $(0, 5, 0, 0, \dots, 0)$ with $n = 10$, max $= 5$, LSE $= 5 + \log(1 + 9e^{-5}) \approx 5.061$. Tight at large margin, off by up to $\log n \approx 2.3$ at zero margin.

2.5 Mechanics check

- **Exponential** — converts any real vector to positive entries, enforcing non-negativity.
- **Normalizer** $Z = \sum_j e^{s_j}$ — enforces sum-to-one, turning a positive-valued map into a probability distribution.
- **Shift-invariance** — makes the implementation numerically stable (subtract the max), and means softmax only cares about score differences, not absolute scales.
- **Temperature T (or equivalently, pre-softmax scaling)** — the only knob controlling concentration. Every selectivity argument in attention is ultimately a temperature argument.
- **Not hard-max** — softmax's smoothness is what makes attention trainable (Prop 1.4 + Lemma 3.3). Argmax would give identical forward-pass semantics in the $T \to 0$ limit but zero gradient.
- **Asymptotics at $T \to 0$ and $T \to \infty$** — one-hot and uniform respectively. The entire useful range of attention lives between these two extremes.

2.6 Exercises

2.1 ⋆⋆ [Mechanism dissection — variance of the categorical.] Let $p = \text{softmax}(s)$ and let X be the categorical random variable with $P(X = i) = p_i$. For any function $f : \{1, \dots, n\} \to \mathbb{R}$, compute $\text{Var}_p(f(X))$ in terms of p and f. Show that the softmax Jacobian (Prop 2.3) is precisely the covariance matrix of the indicator vector $\mathbf{1}_X$ of this categorical. What does this say about the Jacobian's eigenvalues?

2.2 ⋆⋆ [Failure-mode construction — temperature pathologies.] Construct pre-softmax inputs $s^{(1)}, s^{(2)} \in \mathbb{R}^n$ such that:

(a) at $T = 1$, softmax$(s^{(1)})$ and softmax$(s^{(2)})$ have the same top-1 probability, but at $T = 10$ they differ in top-1 by more than 0.1;

(b) the entropy of softmax$_T(s)$ is non-monotone in T over some range.

For (b), state whether non-monotonicity is even possible. Prove or disprove.

2.3 $\star\star\star$ [Theorem about the optimization — LSE duality.] Consider the variational characterization

$$\mathrm{LSE}(s) = \max_{p \in \Delta^{n-1}} \left\{ \langle p, s \rangle + H(p) \right\},$$

where Δ^{n-1} is the probability simplex and $H(p) = -\sum_i p_i \log p_i$.

(a) Prove the identity: show the maximizer is $p = \mathrm{softmax}(s)$ and the maximum value is $\mathrm{LSE}(s)$.

(b) Generalize to temperature: express $T \cdot \mathrm{LSE}(s/T)$ as a similar variational problem with the entropy term scaled appropriately. What does temperature correspond to in this dual picture?

(c) Interpret the duality in attention: what is the "entropy-regularized" optimization that softmax attention is implicitly solving at each query position?

2.4 $\star\star\star$ [Failure mode — hard attention and gradient estimators.] A hard-attention layer selects one key per query via argmax instead of softmax. Gradients vanish almost everywhere (Prop 1.4 failure).

(a) Sketch the Gumbel-softmax estimator: add Gumbel noise to the scores, softmax at temperature T, anneal T toward zero during training. Why does this preserve gradients while approaching hard selection?

(b) Sketch the straight-through estimator: forward pass is argmax; backward pass pretends the argmax was the identity map (or, more commonly, pretends it was softmax). Why is this biased in the gradient sense, and what is the bias? Name at least one failure mode in terms of training dynamics.

(c) Compare: under what circumstances would you choose Gumbel-softmax over straight-through, and vice versa? What role does the scale of n (number of keys) play?

2.5 $\star\star\star$ [Cross-chapter combination — predict-then-verify.] The notebook will compute softmax entropy of $\mathrm{softmax}(\mathbf{s})$ for $\mathbf{s} \sim \mathcal{N}(0, \sigma^2 I_n)$ as a function of (σ, n). Before running, predict:

(a) for fixed n, entropy as a function of σ — direction, asymptotes at $\sigma \to 0$ and $\sigma \to \infty$, inflection behavior;

(b) for fixed σ, entropy as a function of n — direction, scaling with n;

(c) is there a combined (σ, n)-quantity that controls entropy? Conjecture one, defend it, and state what you expect to see in a heatmap of entropy over (σ, n). Hint: the score variance of a single element is σ^2; relate this to Prop 2.5.

State confidence per part with a one-line reason.

2.7 Before the notebook (`02_softmax_temperature.ipynb`)

Write down the following predictions before opening the notebook.

1. **Shift invariance check.** Direction and expected numerical error of the difference $\mathrm{softmax}((1, 2, 3)) - \mathrm{softmax}((101, 102, 103))$ under the safe (max-subtracting) and naive (no subtraction) implementations.
2. **Entropy vs. temperature.** Fix $s \sim \mathcal{N}(0, I_{50})$. Plot of $H(\mathrm{softmax}_T(s))$ as T ranges over $[0.01, 100]$ on a log-x axis. State the two asymptotes and the direction of the curve between them.
3. **Saturation margin.** For $n = 10$ and target $p_{\text{win}} = 0.99$, predict $\Delta^\star$ and the closed-form dependence on n (from Corollary 2.5.1).

4. **Entropy heatmap over (σ, n).** From Exercise 2.5(c): what combined quantity do you conjecture controls entropy? Predict the level-set shape of the entropy heatmap over $\log \sigma \times \log n$ — are level sets vertical, horizontal, or diagonal lines?
5. **Jacobian PSD check.** Softmax Jacobian is a covariance matrix (Exercise 2.1). Predict: is it PSD or strictly PD? What is its rank?
6. **LSE sandwich.** For $s \sim \mathcal{N}(0, I_n)$, predict the gap $\mathrm{LSE}(s) - \max_i s_i$ as a function of n. Does it grow like $\log n$ (the worst-case bound) or something smaller?

For each: direction, magnitude, confidence. The notebook will confirm or update each prediction.